Intro to Spark DataFrame API

What is a DataFrame?

- DataFrame is a structured data type in Spark that can lead better expressivity, performance and space efficiency.
- Dataset is the other structured data type in Spark that is also statically typed (so it is only available in Scala and Java).
- The focus is on *"what to do?"* instead of *"how to do?"*
- The data is structured as a table (similar to a table in a relational database). We can formally describe the schema either directly in the host language or via a Data Definition Language (DDL).
- Simple example to motivate the use of DataFrames: DataFrame Schemas RDDs
- Data types supported in DataFrame in Spark.
- A RDD can be easily converted into a `DataFrame` and vice versa but there is a cost in the transformation.

Examples

- Basic examples:
  - Example 1: DataFrame Schemas
  - Example 2: Working with Columns
  - Example 3: Working with Rows
  - Example 4: Wordcount with DataFrames
- Two complete examples of using data frames.
  - Example 5: Distribution of M&Ms by State
  - Example 6: Fire Call Analytics for San Francisco

DataFrames/Datasets vs RDDs (1))

- If you want to tell Spark "*what to do*" versus "*how to do*", use DataFrames or Datasets.
- If you want high-level abstractions, and domain specific language operators, use DataFrames or Datasets.
- If you want static compile-time safety and don't mind multiple classes for specific Dataset[T], use Datasets (you would have to use Scala or Java).
- If your processing demands high-level expressions, filters, maps, aggregations, computing averages or sums, SQL queries, columnar access, or use of relational operators on semi-structured data, use DataFrames or Datasets.
- If you want to take advantage of and benefit from efficient serialization, use Datasets.

DataFrames/Datasets vs RDDs (2)

- If you are a Python user, use DataFrames and drop down to RDDs if you need more control.
- If you are an R user, use DataFrames.
- If you want errors caught during compilation rather than runtime, choose the appropriate API as shown below.

	SQL	DataFrames	Datasets
Syntax Errors	Runtime	Compile Time	Compile Time
Analysis Errors	Runtime	Run Time	Compile Time

- RDDs remain the fundamental data structure that underlies DataFrames and Datasets. However, all development should use DataFrame (and Dataset) interface.
- We do, sometimes, need to use RDDs if third party package is written in RDDs. Some operators are only available on RDDs.